

Karl Pierce, PhD

Postdoctoral Research Scientist



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LinkedIn

GitHub

Website

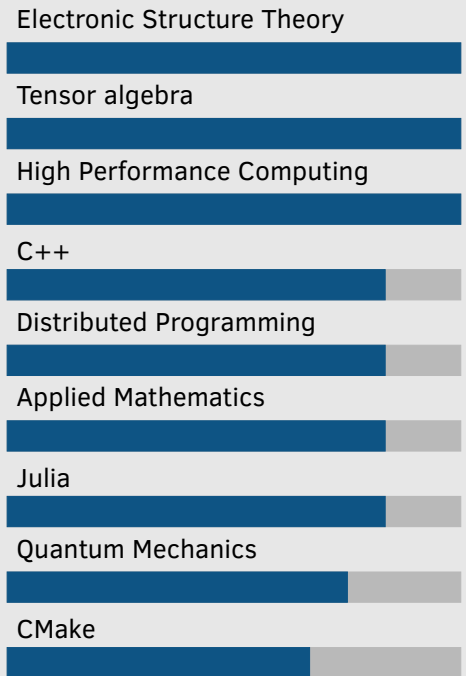


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About Me

Results-driven, adaptable scientist with experience in both laboratory and theoretical research. Thrives in independent and group settings with an ambition to collaborate and improve interdisciplinary research.

Skills



Independent Research★
Communication★ Experimental Design★
Library and Algorithm Design★

Objective

Passionate about the application of emerging tensor algebra to physics, chemistry and other computational modeling problems. Dedicated to creating advanced software tools to improve and accelerate the study of applied mathematics on modern distributed computer infrastructures. Invested in the effective mentoring of students at all levels.

Education

2021	PhD Theoretical Chemistry <i>Virginia Tech</i>	Blacksburg, VA
	Dissertation: Breaking the curse of dimensionality in electronic structure methods: towards optimal utilization of the canonical polyadic decomposition.	
2016	B.S. Chemical Physics <i>Rice University</i>	Houston, TX

Professional Experience

Sept 2025 -	Postdoctoral Associate	The University of Maryland
	Postdoctoral Associate working under Dr. Haizhao Yang (Associate Professor of Mathematics and Computer Science) and Dr. Yuehaw Khoo (Assistant Professor of Statistics at the University of Chicago) studying the application of tensor algebra to machine learning based solutions of partial differential equation problems.	
2022- 2025	Flatiron Software Research Fellow	The Flatiron Institute
	Postdoctoral researcher at the Center for Computational Quantum Physics. Worked with faculty researchers to develop and advance theories in computational quantum physics and chemistry using novel ideas in multilinear algebra. Worked as a lead developer for the ITensors.jl software package. In this package, I created support for heterogenous CPU/GPU algorithms via the use of state-of-the-art generic programming techniques in the Julia programming language.	
2022	Research Scientist	Virginia Tech
	Worked on developing efficient electronic structure methods for large molecules. Built parallel implementations of these methods using the C++ language for heterogeneous massively parallel computer systems. Planned and conducted research projects requiring independent evaluation, selection, and substantial adaptation or modification from standard published techniques and procedures. Furthermore, I mentored graduate students and postdocs.	
2016-2021	Graduate Research Assistant	Virginia Tech
	PhD supervised under Dr. Edward Valeev Studied electronic structure theory, higher-order tensor algebra, and advanced data compression and algorithmic optimization schemes. Developed production level tools in the software packages BTAS , a higher-order tensor algebra library, TiledAr-ray , a scalable tensor framework for high-performance tensor arithmetic, and MPQC , a platform for ab initio electronic structure methods simulation.	
2015-2016	Research Assistant	Rice University
	Supervised under Dr. Gustavo Scuseria	
2014-2015	Research Assistant	Rice University
	Supervised under Dr. Emilia Morosan	
2013	Visiting Scientist	University of Akron
	Supervised under Dr. Shing-Chung "Josh" Wong	

Publications

2025	Towards Using Matrix-Free Tensor Decomposition to Systematically Improve Approximate Tensor-Networks. <u>Pierce, K</u> J. Chem. Theory Comput. 2025, 21, 13, 6464–6481
2025	Using Matrix-Free Tensor-Network Optimizations to Construct a Reduced-Scaling and Robust Second-Order Møller-Plesset Theory. <u>Pierce, K</u> ; Morales, M J. Chem. Theory Comput. 2025, 21, 12, 5952–5964
2022	Efficient Construction of Canonical Polyadic approximations of Tensor Networks <u>Pierce, K</u> ; Valeev, E. J. Chem. Theory Comput. 2023, 19, 1, 71–81.
2021	Robust Approximation of Tensor Networks: Application to Grid-Free Tensor Factorization of the Coulomb Interaction. <u>Pierce, K</u> ; Rishi, V; Valeev, E. F. J. Chem. Theory Comput. 2021, 17, 4, 2217–2230.
2020	Massively Parallel Quantum Chemistry: A high-performance research platform for electronic structure. Peng C; Lewis C; Xiao W; Clement M; <u>Pierce, K</u> ; Rishi V; Pavošević F; Slattery S; Zhang J; Teke N; Kumar A; Masteran C; Asadchev A; Calvin J; Valeev E. F; (2020). J. Chem. Phys., 153(4), 44120.
Accepted	STTID: High-Performance Sparse Tensor-Train Interpolative Decomposition Z. Meng; EM. Stoudenmire; <u>Pierce, K</u> ; F. Mueller; J. Li
Submitted	A CPD-enabled low-scaling environment solver in a coupled cluster based static quantum embedding theory <u>Pierce, K</u> ; Aziz, M; Shee; FM Faulstich; (2026) arXiv:2602.15129v1
In Preparation	Accelerating the Canonical Polyadic Alternating Least Squares Optimization via a Randomized Interpolative Decomposition Fakih, I; Grigori, L; <u>Pierce, K</u> Effective use of 4-way Canonical Polyadic Decomposition for Accelerating the Coupled-Cluster Perturbative Triples. <u>Pierce, K</u> ; Bao, Y; Pavošević, F; Valeev, E Studying the Efficient Approximation of Lattice Networks using the Canonical Polyadic Decomposition <u>Pierce, K</u> ; Tindall, J; Stoudenmire EM Blocked Leverage Scores in Randomized Least Squares for CP Tensor Decomposition <u>Pierce, K</u> ; Bellavita, J; Liu, Y; Nahar Keya, K; Haddock, J; Kolda, T The Prediction of Accurate Ground State Energies via Rank-Reduced Periodic Coupled Cluster and the Interpolative Separable Density Fitting Approximation. Pimpel, F; Morales, M; Grüneis, A; <u>Pierce, K</u> GPU Performance of Dense and Sparse Tensor Algorithms using the ITensor Library <u>Pierce, K</u> ; Fishman, M; EM Stoudenmire An Efficient Application of the Canonical Polyadic Decomposition to the Rank-Reduced Coupled Cluster Method <u>Pierce, K</u>

Invited Presentations and Posters

2026	Workshop: Tensor Decompositions in Polynomially-scaling Many-body Expansion Methods The Canonical Polyadic Decomposition of Two-Electron Integrals in Quantum Chemistry	Seminar
2026	27th Conference of the International Linear Algebra Society Blocked Leverage Score Sampling in the Randomized Least Squares for CP Tensor Decomposition	Seminar
2026	SIAM Conference on Parallel Processing for Scientific Computing Accelerating the Canonical Polyadic Decomposition via a Randomized Interpolative Sampling Procedure	Seminar

2025	Molecular Quantum Mechanics Conference Application of the Canonical Polyadic Decomposition in the Pursuit of Extremely Low-Scaling Correlation Methods.	Seminar
2025	NVIDIA Design, Demos, Ideas, and Research discussions Studying the effective application of low-rank tensor approximations to electronic structure methods.	Seminar
2025	SIAM Conference on Computational Science and Engineering Towards the Efficient Approximation of Higher-Order, Tensor-Network Contractions Via a Low-Rank, Matrix-Free Tensor Decomposition	Seminar
2025	Applied Math Colloquim at Rensselaer Polytechnic Institute Investigating the Utility of Matrix-Free Tensor Decompositions in Quantum Simulations.	Seminar
2024	Juliacon Improving the life-cycle of tensor algorithm development	Seminar
2024	Invited group meeting: Virginia Tech Approximating tensor contractions via a matrix-free tensor decomposition	Seminar
2024	Extreme-scale Mathematically-based Computational Chemistry meeting Approximating tensor contractions via a matrix-free tensor decomposition	Seminar
2024	Invited group meeting: Vienna University of Technology Approximating tensor contractions via a matrix-free tensor decomposition	Seminar
2023	SIAM Conference on Computational Science and Engineering Introduction to the ITensor Software Library for Tensor Network Calculations	Seminar
2021	Colloquim at Vienna University of Technology <i>Utility of the Canonical Polyadic Decomposition and Robust Tensor Network Approximations</i>	Seminar
2019	Virginia Tech Department of Chemistry Internal Seminar <i>Reduced Cost Electronic Structure Theory via the Canonical Polyadic Decomposition</i>	Seminar
2019	American Chemical Society National Meeting <i>Towards Reduced Scaling Higher Order Coupled Cluster Methods via Tensor Decomposition.</i>	Poster
2018	Modern Wavefunction Methods in Electronic Structure Theory <i>Reducing Complexity and Cost of High-Order Coupled-Cluster Method via Canonical Polyadic Decomposition of Hamiltonian</i>	Poster
2018	Penn Conference in Theoretical Chemistry and Electronic Structure Workshop <i>Reducing Complexity and Cost of High-Order Coupled-Cluster Method via Canonical Polyadic Decomposition of Hamiltonian.</i>	Poster
2018	Virginia Tech Department of Chemistry Preliminary Exam <i>Reduced Scaling of Accurate Electronic Structure Methods using Tensor Decompositions</i>	Seminar
2017	Southeast Theoretical Chemistry Association Meeting <i>Toward Efficient Canonical Product Decomposition in TiledArray Framework</i>	Poster

Workshops and Advanced Topic Schools

2025	Research Collaboration Workshop, "Randomized Numerical Linear Algebra" (RNLA) Research Workshop that focuses on developing efficient and practical algorithms for solving problems using numerical linear algebra via randomized techniques. Assigned to work with Dr. Tamara Kolda on a project to study structure-aware randomization for linear algebra.	Los Angeles, CA
2022	Argonne Training Program on Extreme-scale Computing Participated in an intensive two week summer school learning modern key skills, approaches, and about tools to design, implement and execute scientific applications on state of the art, leadership-class computing systems of today and the future.	Chicago, IL
2018	MolSSI Summer School and Workshop Parallel Computing in Molecular Sciences Participated in a three-day lecture series where researchers in academia and from Berkeley national lab. Discussed computational parallelism and communication on homogenous and heterogeneous CPU/GPU computer systems.	Berkely, CA

2018 Modern Wavefunction Methods in Electronic Structure Theory Gelsenkirchen, Germany
Attended a week-long summer school at the Max-Planck institute in Germany directed towards Ph.D. students and postdocs with aims to teach advanced topics in the field of ab initio electronic structure theory, reduced scaling algorithms, and software implementations on modern hardware.

Certificates

2025 LinkedIn Training Neural Networks in Python
2022 NVIDIA Certificate in Scaling CUDA C++ Applications to Multiple Nodes
2022 NVIDIA Certificate in Fundamentals of Accelerated Computing with CUDA C/C++
2025 LinkedIn Training Neural Networks in Python.

Teaching and Mentoring Experience

Course work

University of Maryland

Department of Mathematics: Data Science and Analytics

The Flatiron Institute

Graduate Scholar Workshop Series

qnumerics: School for Numerical Methods in Quantum Information Science

Virginia Tech

Physical Chemistry, Thermodynamics

General Chemistry Lab

Physical Chemistry Lab

Virginia Tech: General Chemistry Lab

Mentoring

Mina Mandic

Flatiron Institute Undergraduate Summer internship. Presented work at SIAM NY/NJ/PA regional meeting 2024.

Florian Pimpel

PhD Candidate at TU Wien. Studying reduced-scaling approximations for periodic coupled cluster.

Muhammad Aziz

PhD Student at Rensselaer Polytechnic Institute. Studying reduced-scaling approximations for chemistry-embedding problems.

Israa Fakh

PhD Student at the Institute of Mathematics, EPFL. Studying the application of randomized linear algebra in tensor decompositions algorithms.

Kamrun Nahar Keya

PhD Student at the Arizona State University. Studying the canonical polyadic decomposition, computer programming fundamentals and application of randomized linear algebra in tensor decompositions algorithms.

Projects

ITensorCPD.jl

Developed an efficient tensor algebra library to support the canonical polyadic decomposition of tensors and tensor networks interfacing with ITensors.jl and ITensorNetworks.jl. Implementing novel algorithms using randomized linear algebra.

ITensors.jl

Developed generic libraries to seamlessly integrate GPU support. Worked with lead developers to reorganize, simplify and extend the functionality of the ITensor and internal Libraries.

BTAS

Developed efficient serial tensor algebra kernels. Implemented existing and novel tensor decomposition algorithms.

TiledArray

Developed distributed tensor algebra kernels using the OMP based task scheduling software **madness**.
Developed an efficient block distributed canonical polyadic decomposition.

MPQC

Developed canonical electronic structure methods such as Hartree-Fock, Møller-Plesset 2 perturbation theory, and various levels of coupled cluster theory. Utilized higher-order decompositions schemes to develop reduced-cost and scaling implementations of canonical electronic structure methods.

Professional Affiliations

The American Chemical Society
Society of Industrial and Applied Mathematicians

Awards

2021	Graduate School Doctoral Assistanship Award Award for excellence in research in leadership	Virginia Tech
2012,2014	C-USA Commissioner's Honor Roll	Rice University

Extra-curricular Activities

2017-2023	Pole Vault Coach Designed individualized athletic training and programming as a head coach for youth athletes.	Blacksburg High School
2012-2016	Division 1 Athlete Participated in Division 1 athletics at Rice University as a pole vaulter on the track and field team.	Rice University